

LDACS1310: Introduction to Applied Cryptography

Lecture 3: block ciphers

T. Peters, ***F.-X. Standaert***

UCLouvain Crypto Group, fstandae@uclouvain.be

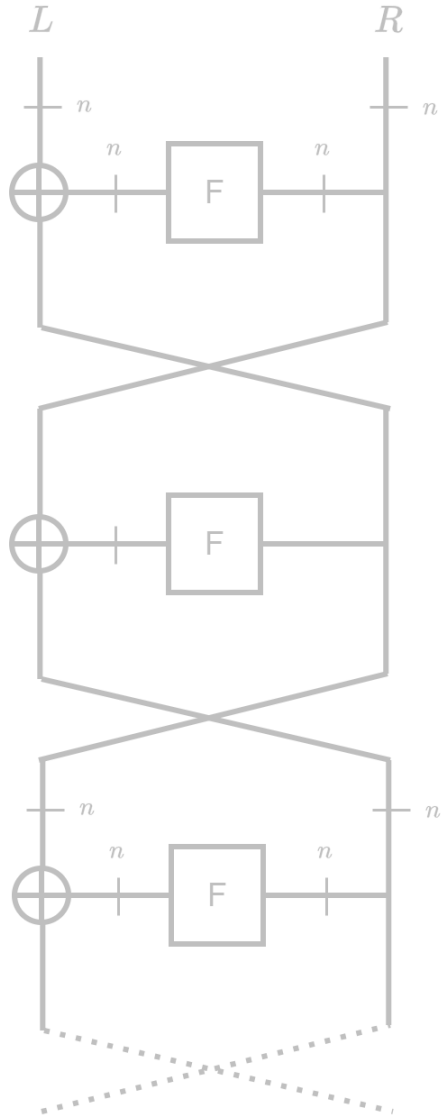
Reminder



<https://perso.uclouvain.be/fstandae/LDACCS1310/>

And questions from last lecture?

Assumptions dilemma: P from F



- $F : \{0,1\}^n \times \{0,1\}^n \rightarrow \{0,1\}^n$
 $\Rightarrow P : \{0,1\}^n \times \{0,1\}^{2n} \rightarrow \{0,1\}^{2n}$

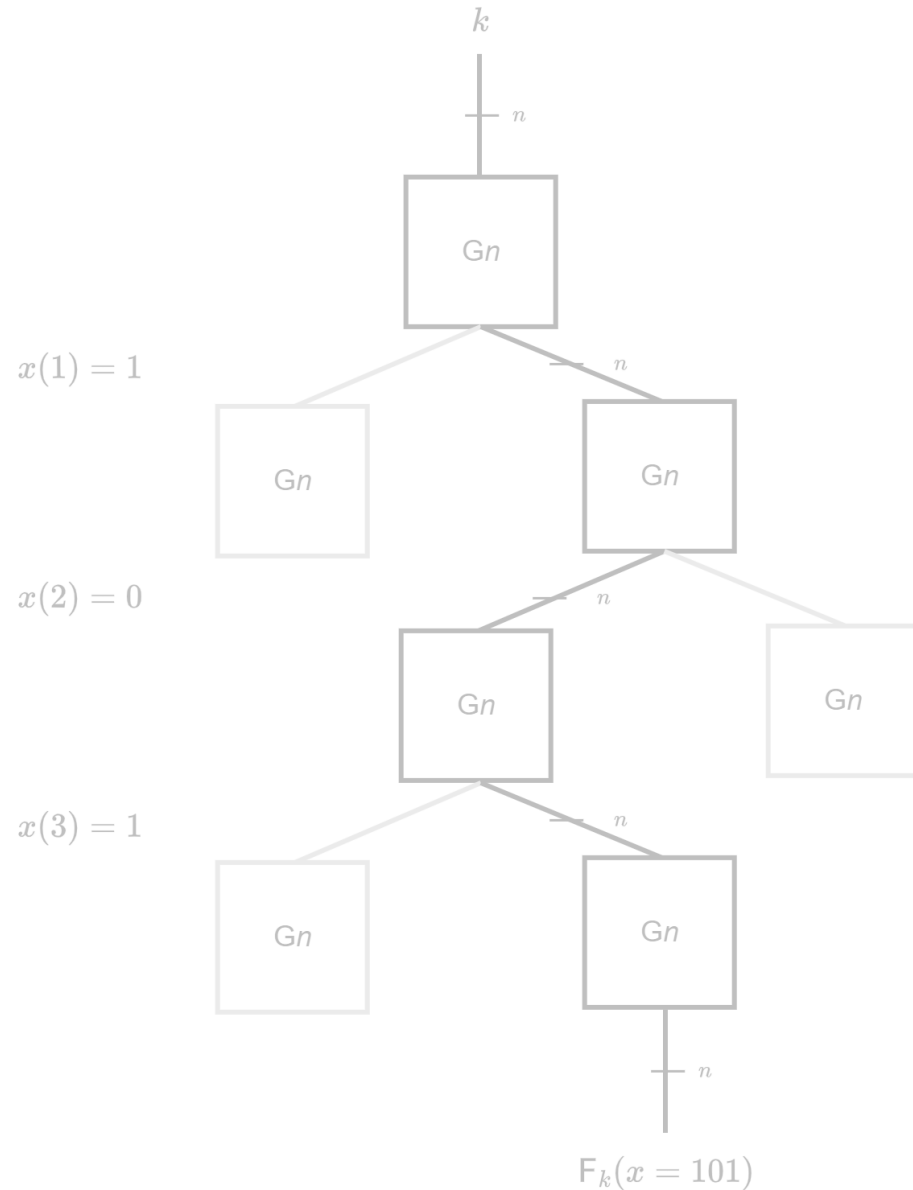
$$3 \text{ rounds: } \epsilon_P \approx \frac{q^2}{2^n} + 3 \cdot \epsilon_F$$

How to invert? Is it tight? {...}

$$7+ \text{ rounds: } \epsilon_P \approx \mathcal{O}(2^{n \cdot (1-\alpha)}) + 3 \cdot \epsilon_F$$

- Reverse direction is trivial (use the PRP as PRF) but $\epsilon_F \approx \frac{q^2}{2^{2n}} + \epsilon_P$

Assumptions dilemma: F from Gn



- $G_n: \{0,1\}^n \rightarrow \{0,1\}^{2n}$
 $\Rightarrow F: \{0,1\}^n \times \{0,1\}^m \rightarrow \{0,1\}^n$

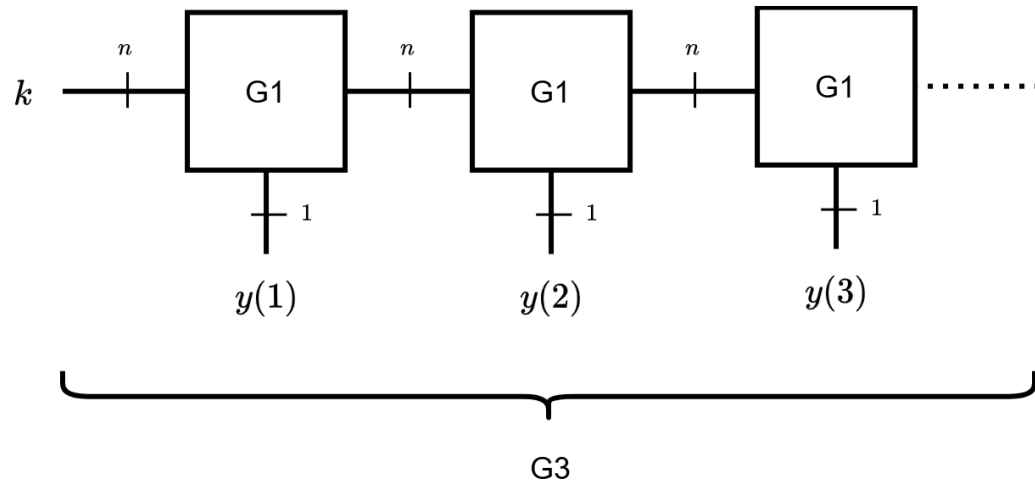
$$\epsilon_F \approx q \cdot m \cdot \epsilon_{G_n}$$

Is it tight? {...}

- Reverse direction is trivial (just use a counter as PRF input)

Assumptions dilemma: G_n from G_1

- $G_1: \{0,1\}^n \rightarrow \{0,1\}^l \Rightarrow G_n: \{0,1\}^n \rightarrow \{0,1\}^{2n}$



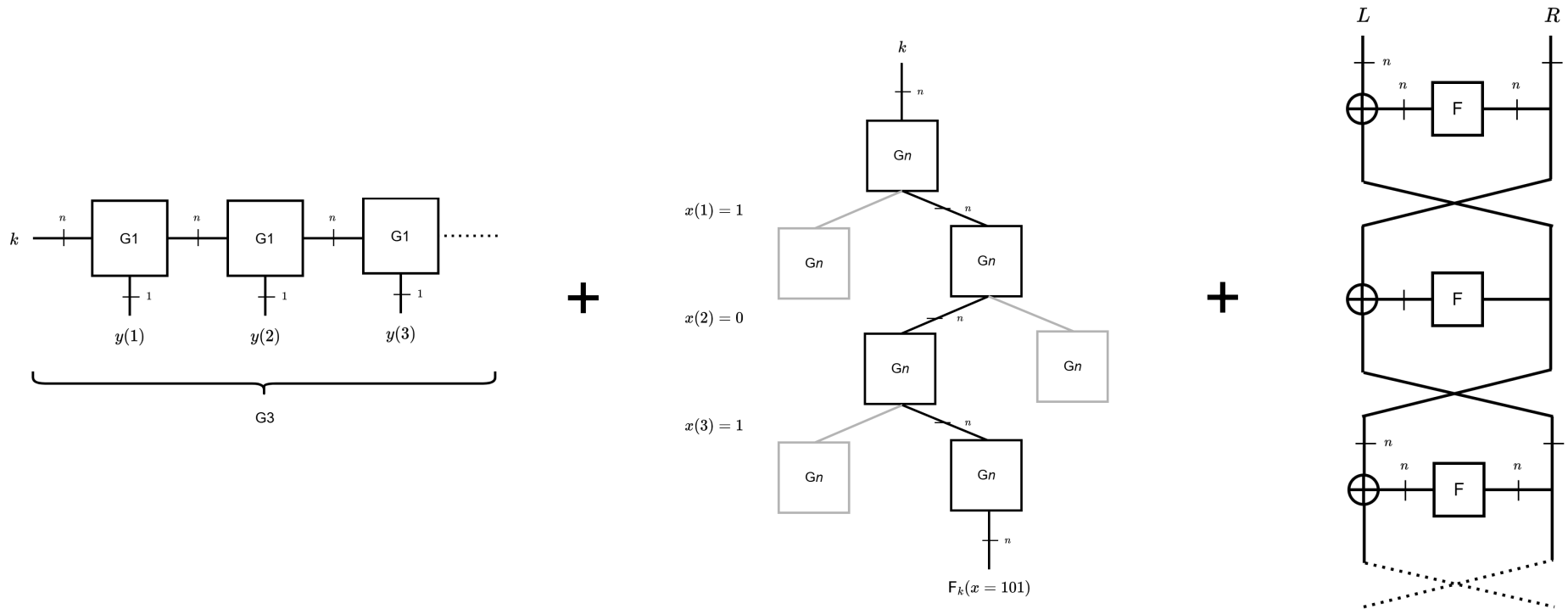
$$\epsilon_{G_n} \approx n \cdot \epsilon_{G_1}$$

Is it tight? {...}

- Reverse direction is trivial (just take one bit out of n)

Efficiency challenge

❑ Question: say $n = 128$, $G1$ takes 1ms, how long for 16 PRP rounds?



⇒ Strong motivation for **direct PRP design** ($\approx$ heuristic shortcut)

Efficiency challenge, revisited

❑ Question: to design a 128-bit PRP, why not picking it at random and storing it as a table?

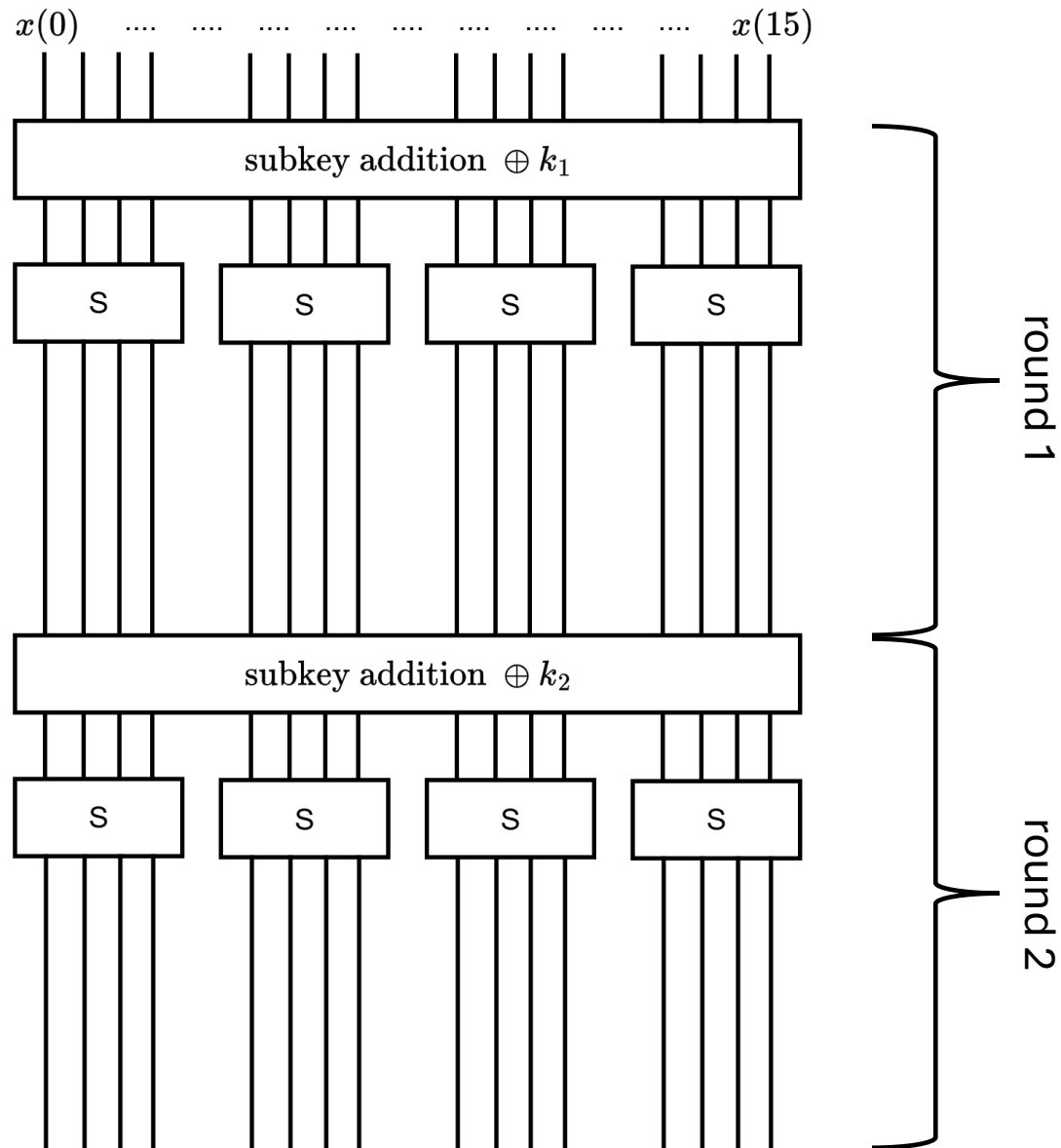
❑ Question: could a n -bit PRP be linear, e.g., $y = A \cdot (x \oplus k)$ with A defined as a $n \times n$ invertible (bit) matrix in $\mathbb{F}_2$?

❑ How many such linear functions are there?

❑ Complexity of solving a system of n linear equations in n variables?

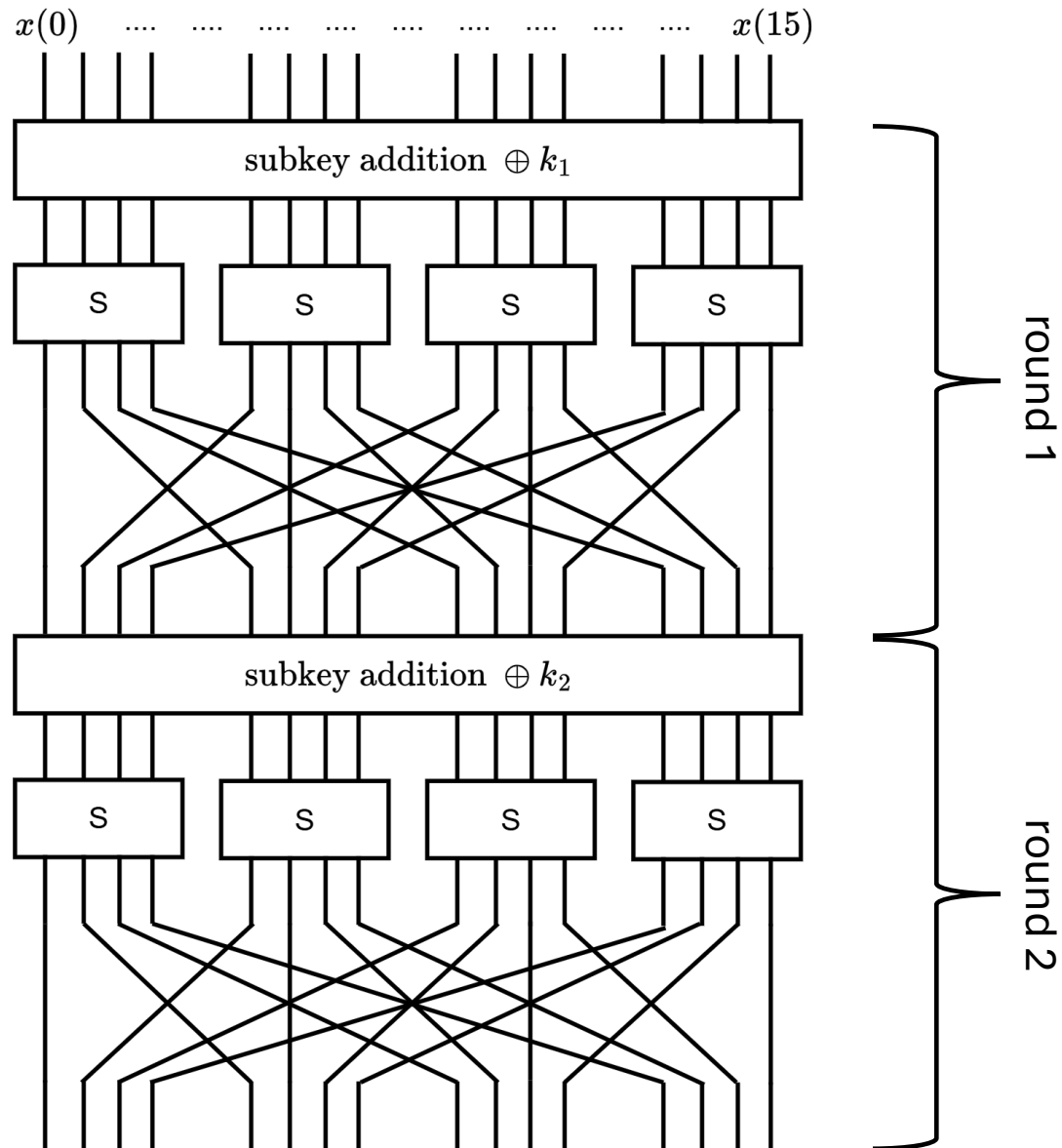
⇒ Block ciphers ($\approx$ efficient PRP) must be non-linear (bring *confusion*)

Key alternating cipher



- ❑ Question: say S is a (small) random permutation (called S-box), what is the complexity of an exhaustive key search against the toy block cipher of the figure?
- Expected:
 - Observed:

Key alternating cipher



❑ Question: say S is a (small) random permutation (called S-box), what is the complexity of an exhaustive key search against the toy block cipher of the figure?

- Expected:
- Observed:

⇒ Block ciphers also need *diffusion*

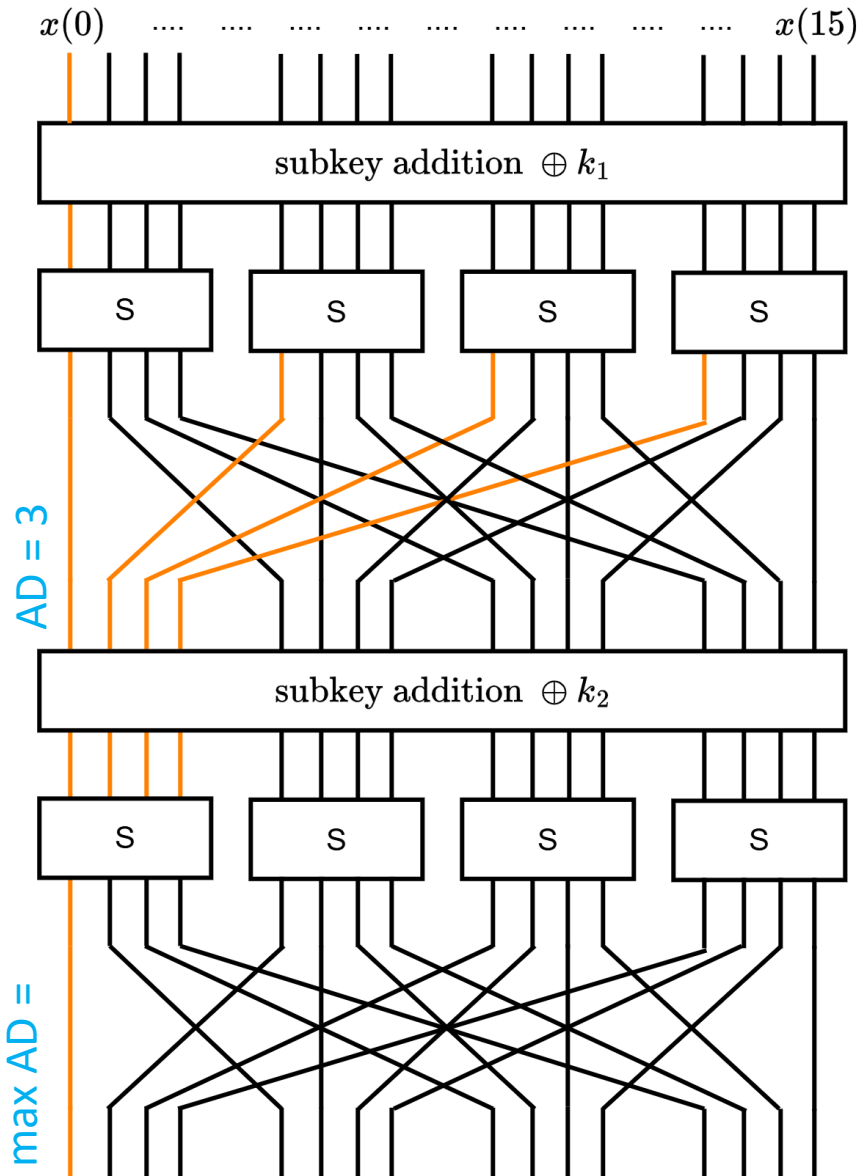
Algebraic cryptanalysis

- Even simpler toy example x and k are 2-bit wide and

$$\begin{array}{ll} y(0) = x(0) \oplus k(0) & y(1) = x(1) \oplus k(1) \\ z(0) = y(0) \cdot y(1) & z(1) = y(1) \oplus y(0) \end{array} \left. \vphantom{\begin{array}{l} y(0) = x(0) \oplus k(0) \\ y(1) = x(1) \oplus k(1) \end{array}} \right\} \begin{array}{l} \text{2-bit key addition} \\ \text{and S-box} \end{array}$$

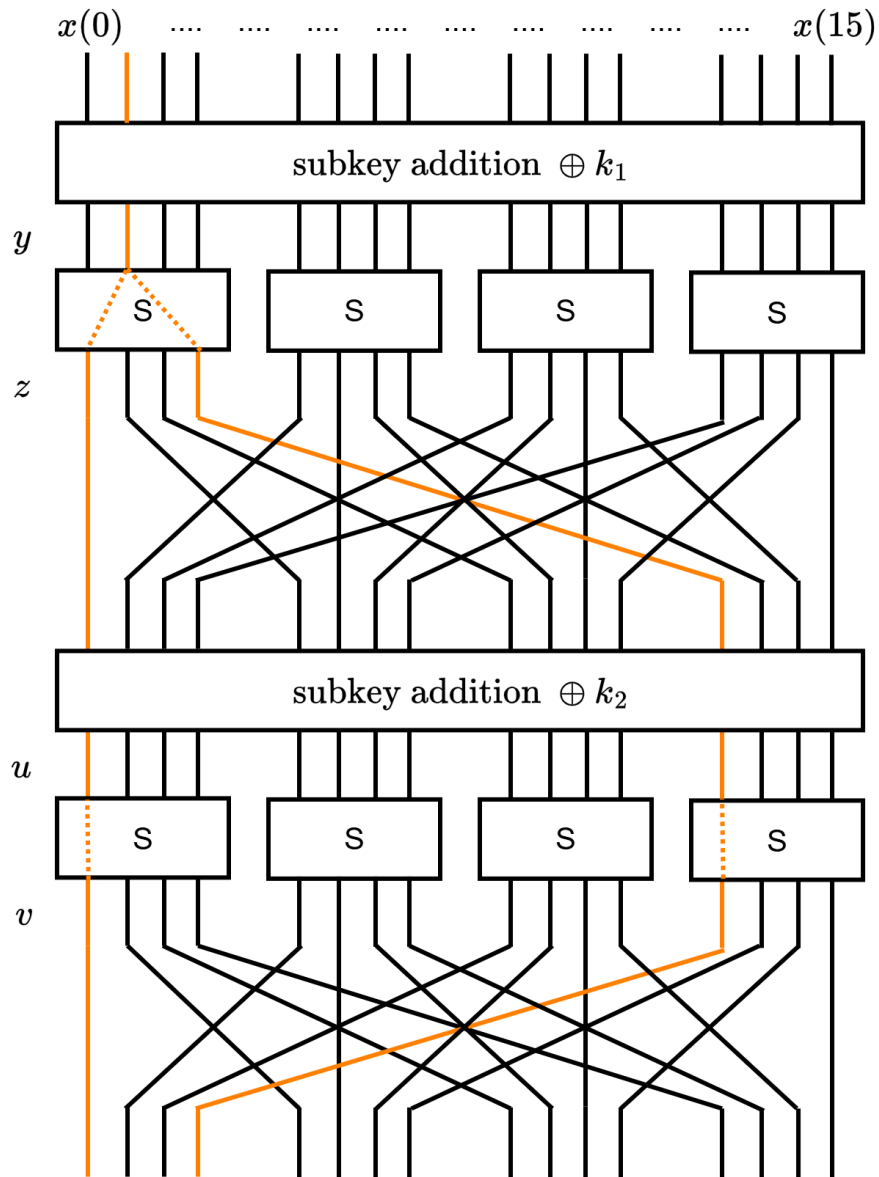
- Leading to
$$z(0) = x(0) \cdot x(1) \oplus x(0) \cdot k(1) \oplus x(1) \cdot k(0) \oplus k(0) \cdot k(1)$$
$$z(1) = x(0) \oplus x(1) \oplus k(0) \oplus k(1)$$
- Quadratic term (algebraic degree 2), can be solved via linearization: replace non-linear terms by extra variables, e.g., $v = k(0) \cdot k(1)$
- Security impact: one more variable & one more linear equation

Security argument



- ❑ Question: say S has optimal algebraic degree (i.e., $d = b - 1$ for a b -bit S-box), meaning $d = 3$ in our example, what is the maximum algebraic degree that can be reached at round 2?
- ❑ Question total/average # of monomials of degree d for a random Boolean function:
 - Linearized system rapidly hard to solve
 - Beware: algebraic degree is only a proxy

Statistical (linear) cryptanalysis (I)



- Say S-box has optimal algebraic degree but
 - $y(1) = z(0) \oplus z(3)$ with proba p_1
 - $u(0) = v(0)$ with proba p_2
 - $u(12) = v(12)$ with proba p_3

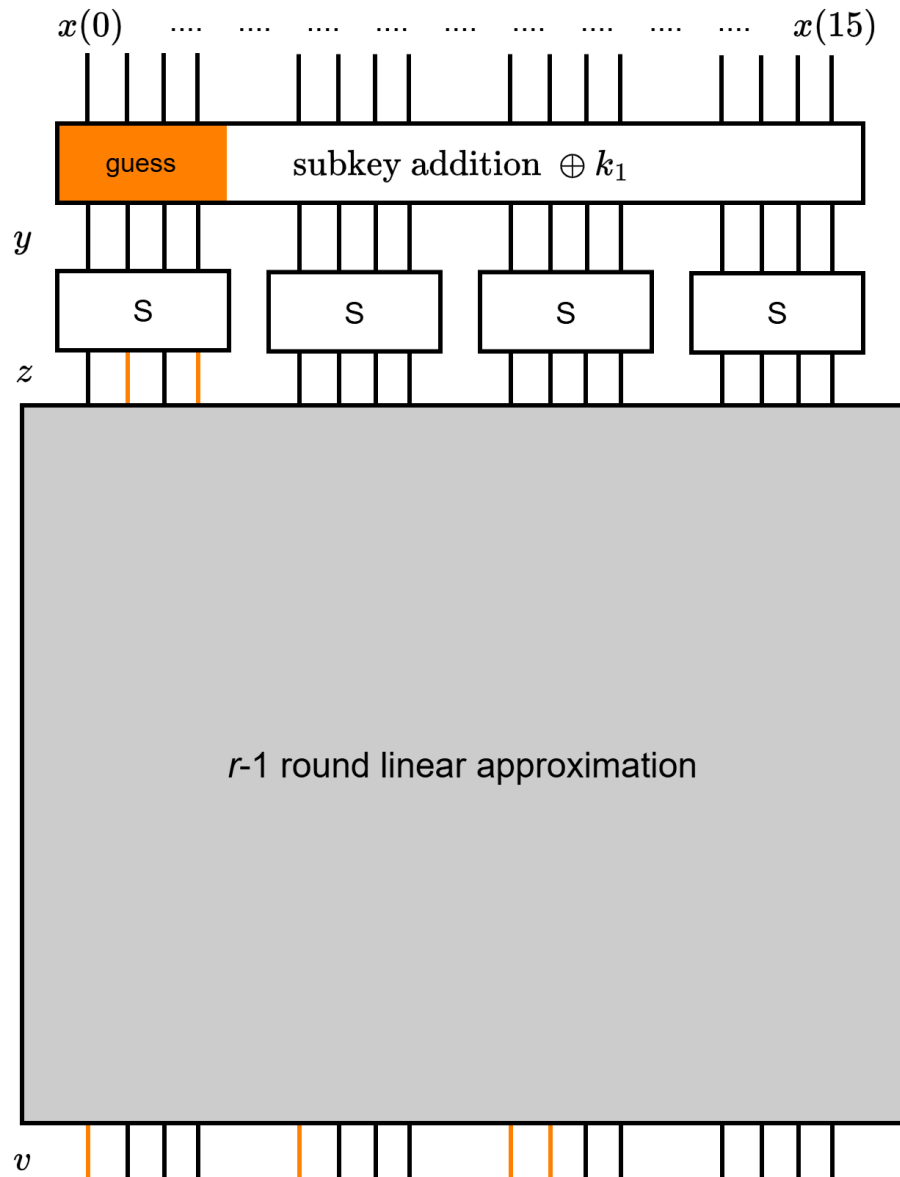
- Define the bias $\epsilon_i = p_i - 1/2$. Then

$$x(1) \oplus k_1(1) = v(0) \oplus k_2(0) \oplus v(12) \oplus k_2(12)$$
 holds with bias $\epsilon = 2^{3-1} \cdot \prod_{i=1}^3 \epsilon_i$

□ Say all $\epsilon_i = 2^{-4}$, then $\epsilon =$
 How does that enable key recovery? {...}

(cf. reading on the course website)

Statistical (linear) cryptanalysis (II)

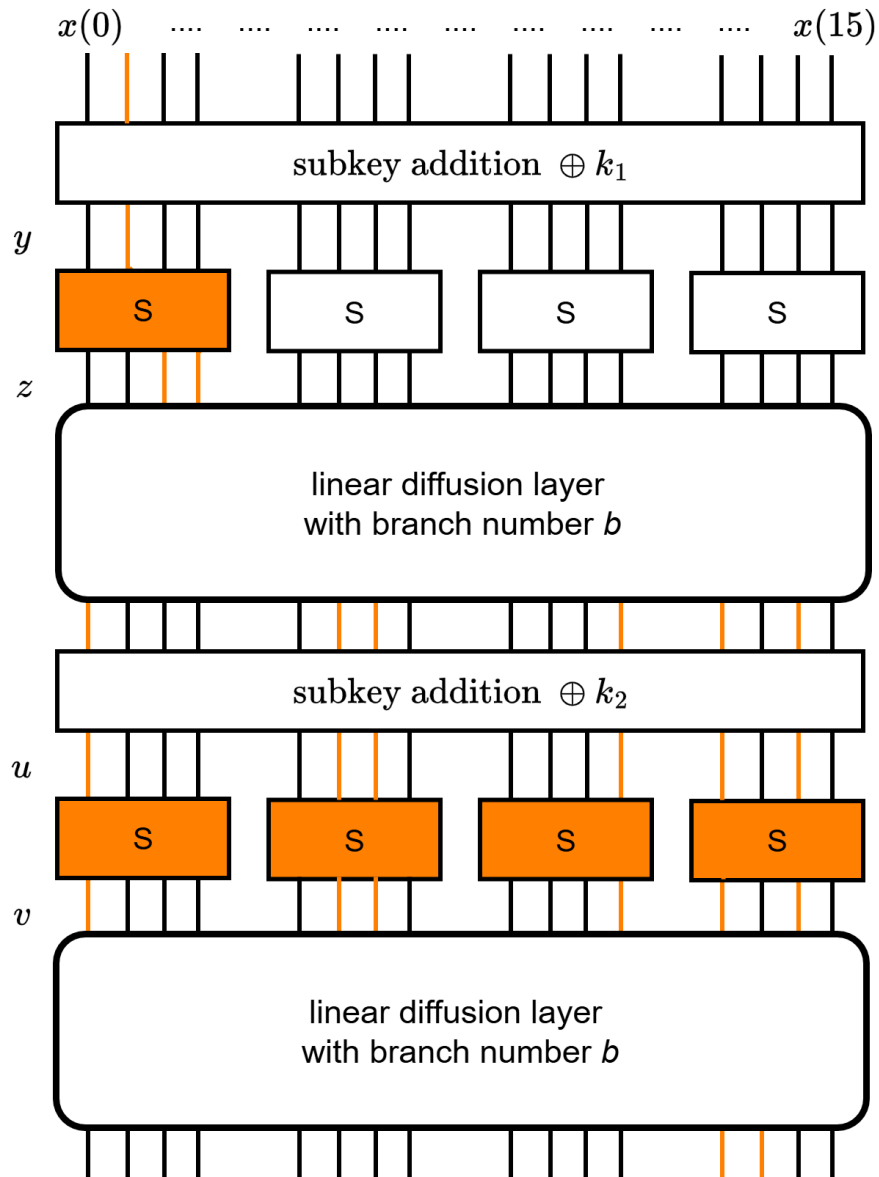


- Use a $r-1$ linear approximation with bias ϵ
- Collect q input / output block cipher pairs
- For all possible $(k_1(0), k_1(1), k_1(2), k_1(3))$, estimate the empirical bias ϵ_{k^*} , that is

$$\widehat{\Pr}(z^*(1) \oplus z^*(3) = v(0) \oplus v(4) \oplus v(8) \oplus v(9)) - 1/2$$
- The bias of the correct key candidate ϵ_{k^*} should be max for $k^* = k$ after $q \approx$
- Repeat for other bits of k_1 (*other approx.*)

(cf. reading on the course website)

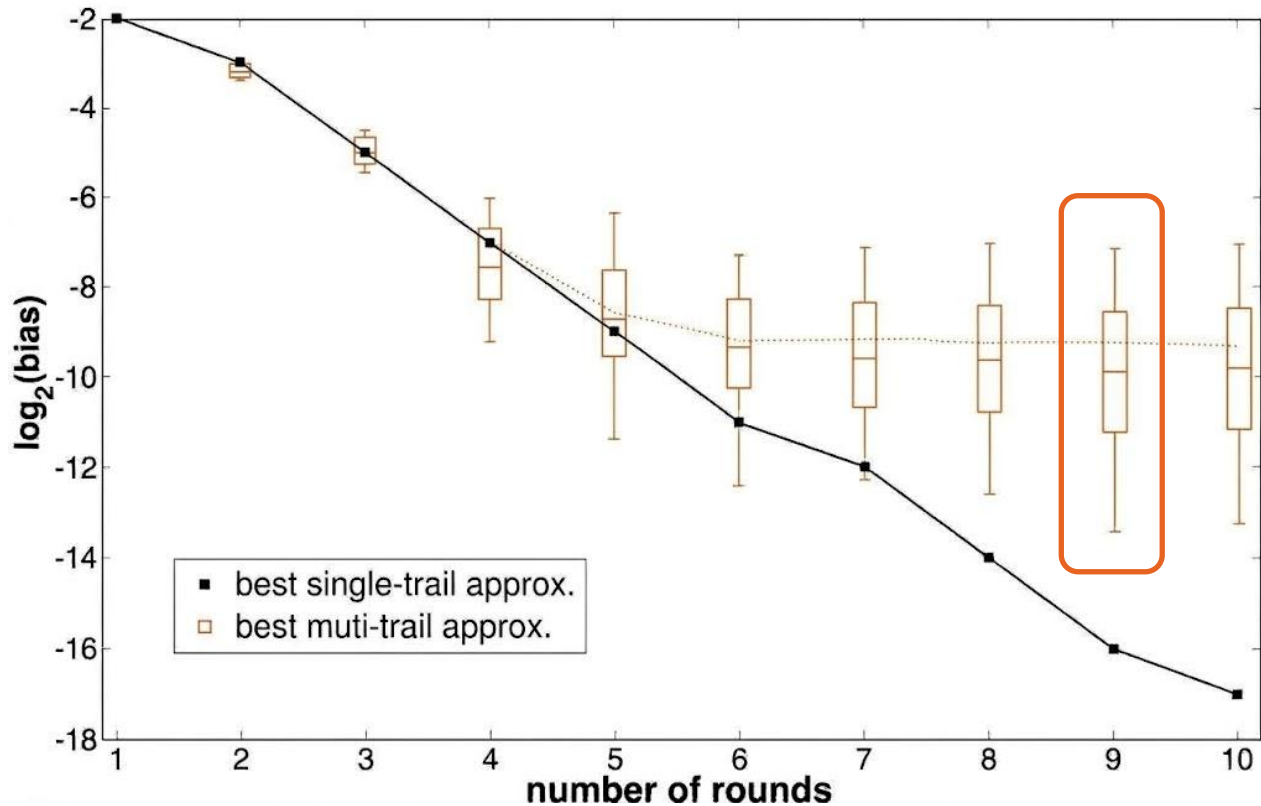
Security argument



- What makes linear cryptanalysis hard?
 -
 -
 -
- Advanced Encryption Standard ([AES](#)) ensures that with “separation of duties”
 - S-boxes with max bias $\epsilon = 2^{-3}$
 - Linear layer with good **branch number** (the sum of the # of active S-boxes at the layer’s inputs and outputs equals b)
 - “Square amplification” with [ShifRows](#) {...}

Discussion

16-bit toy cipher

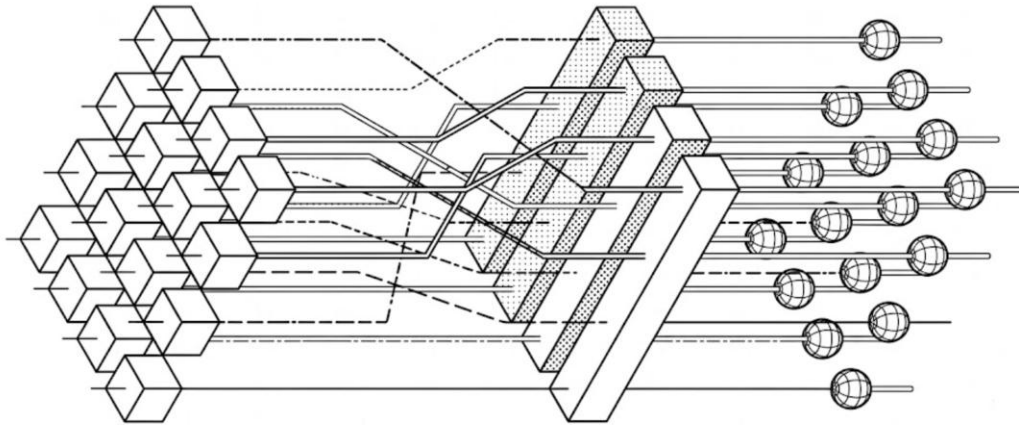


- For low # of rounds, a single linear trail dominates
- For larger # of rounds, many linear trails interfere
- Concretely, it means
 - $\lim_{r \rightarrow \infty} (\epsilon_{\text{single-trail}}) \dots$
 - $\lim_{r \rightarrow \infty} (\epsilon_{\text{multi-trail}}) \dots$

- But multi-trail approx. are hard to exploit (**key-dependent**) so having the squared bias of the best linear approximation $\ll 2^{-n}$ is a good proxy

Summarizing

- AES is standardized as Federal Inf. Process. Stds. (NIST FIPS)-197
- It is widely believed to resist cryptanalysis (20+ years effort)
 - This includes algebraic & linear but also **many others**



- AES operations are carried in binary fields (for efficiency)
 - It offers excellent performances in software and hardware

Towards composition challenges

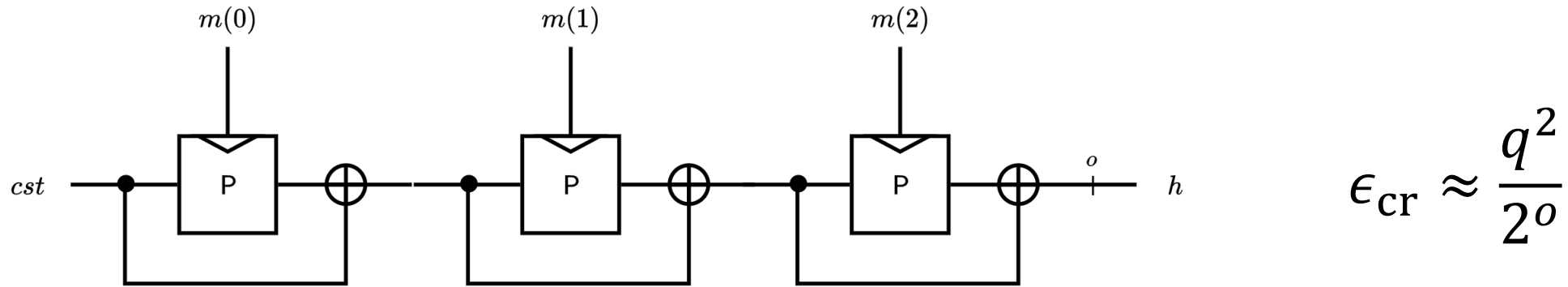
- **Quantum threat:** Grover's algorithm theoretically allows quadratic speedup for exhaustive key search ($\approx$ the need of 2^{128} classical operations is reduced to a need of 2^{64} quantum operations)
 - One needs to double the key size to cover this risk
 - Question, can we use $\text{AES}_k(\text{AES}_{k'}(x))$ for this purpose?
 - Expected (classical) complexity:
 - Observed complexity:
- ⇒ Extending a cipher's domain is not trivial (remember P from F) and motivates ad hoc designs (e.g., AES-256 is a good PRP)

Another useful building block

- Collision-resistant hash functions $H : \{0,1\}^* \times \{0,1\}^o$

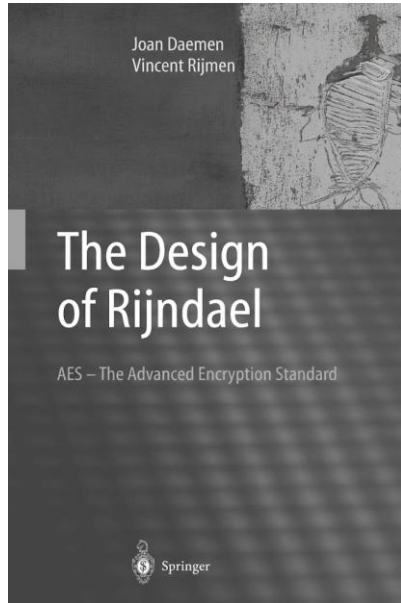
$$\Pr(\text{Adv}^{H(\cdot)} \rightarrow (m_0, m_1 \text{ s. t. } H(m_0) = H(m_1))) \leq \text{negl}(n)$$

- Can be constructed from a block cipher



- Beware: “key” controlled by the Adv $\Rightarrow$ PRP security is not enough)
- Ad hoc designs like the **SHA3** standard are usually more efficient

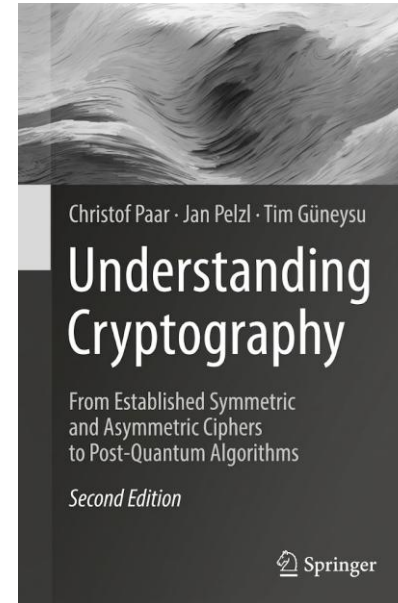
References



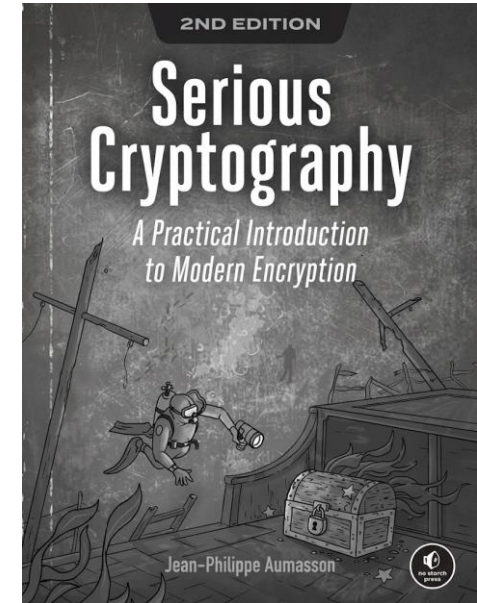
(a)



(b)



(c)



(d)

This lecture builds on the content of References (a), (b) and (c), which provide a detailed and complementary view of the design principles of modern block ciphers. Reference (a) additionally provides a gentle introduction to finite fields, which is useful to understand the internal operations of the AES, on which it focuses. Reference (d) provides discusses these topics in a less technical manner and provides a simple explanation and discussion of padding oracle attacks.

Questions?